

CHAPTER 4

The p-Block Elements (Group 15 to 18)

Group 15 Elements (Nitrogen family)

- Identify the incorrect statement. (2024 Re)
 - PEt_3 and AsPh_3 as ligands can form $d\pi-d\pi$ bond with transition metals
 - The N – N single bond is as strong as the P – P single bond
 - Nitrogen has unique ability to form $p\pi-p\pi$ multiple bonds with nitrogen, carbon and oxygen
 - Nitrogen cannot form $d\pi-p\pi$ bond as other heavier elements of its group
- Which of the following reactions is a part of the large scale industrial preparation of nitric acid (2022 Re)
 - $\text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O} \xrightarrow[500\text{ K, 9 bar}]{\text{Pt}} 4\text{HNO}_3 + \text{Cu}$
 - $\text{NaNO}_3 + \text{H}_2\text{SO}_4 \xrightarrow[500\text{ K, 9 bar}]{\text{Pt}} \text{NaHSO}_4 + \text{HNO}_3$
 - $4\text{NH}_3 + 5\text{O}_2 \text{ (from air)} \xrightarrow[500\text{ K, 9 bar}]{\text{Pt}} 4\text{NO} + 6\text{H}_2\text{O}$
 - $4\text{HPO}_3 + 2\text{N}_2\text{O}_5 \xrightarrow[500\text{ K, 9 bar}]{\text{Pt}} 4\text{HNO}_3 + \text{P}_4\text{O}_{10}$
- Urea reacts with water to form A which will decompose to form B. B when passed through Cu^{2+} (aq), deep blue colour solution C is formed. What is the formula of C from the following? (2020)
 - $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 - $\text{Cu}(\text{OH})_2$
 - $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
 - CuSO_4
- Among the compounds shown below which one revealed a linear structure? (2020-Covid)
 - HOCl
 - O_3
 - N_2O
 - NO_2
- Match the following: (2019)

Column-I		Column-II	
A.	Pure nitrogen	i.	Chlorine
B.	Haber process	ii.	Sulphuric acid
C.	Contact process	iii.	Ammonia
D.	Deacon's process	iv.	Sodium azide or Barium azide

Which of the following is the correct option?

- | | | | |
|--------|-----|-----|-----|
| A | B | C | D |
| a. i | ii | iii | iv |
| b. ii | iv | i | iii |
| c. iii | iv | ii | i |
| d. iv | iii | ii | i |

- The correct order of N-compounds in its decreasing order of oxidation states is (2018)
 - $\text{HNO}_3, \text{NO}, \text{N}_2, \text{NH}_4\text{Cl}$
 - $\text{HNO}_3, \text{NO}, \text{NH}_4\text{Cl}, \text{N}_2$
 - $\text{NH}_4\text{Cl}, \text{N}_2, \text{NO}, \text{HNO}_3$
 - $\text{HNO}_3, \text{NH}_4\text{Cl}, \text{NO}, \text{N}_2$
- Which is the correct statement for the given acids? (2016 - I)
 - Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid.
 - Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid
 - Both are diprotic acids
 - Both are triprotic acids
- When copper is heated with conc. HNO_3 it produces: (2016 - I)
 - $\text{Cu}(\text{NO}_3)_2$ and N_2O
 - $\text{Cu}(\text{NO}_3)_2$ and NO_2
 - $\text{Cu}(\text{NO}_3)_2$ and NO
 - $\text{Cu}(\text{NO}_3)_2, \text{NO}$ and NO_2
- Strong reducing behaviour of H_3PO_2 is due to: (2015 Re)
 - Presence of two –OH groups and one P – H bond
 - Presence of one –OH group and two P – H bonds
 - High electron gain enthalpy of phosphorus
 - High oxidation state of phosphorus
- Nitrogen dioxide and sulphur dioxide have some properties in common. Which property is shown by one of these compounds, but not by the other? (2015)
 - Is soluble in water
 - Is used as a food preservative
 - Forms 'acid-rain'
 - Is a reducing agent
- Which of the following does not give oxygen on heating? (2013)
 - KClO_3
 - $\text{Zn}(\text{ClO}_3)_2$
 - $\text{K}_2\text{Cr}_2\text{O}_7$
 - $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

Group 16 Elements (Oxygen Family)

- Given below are two statements: (2024)

Statement I: The boiling point of hydrides of Group 16 elements follow the order
 $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$.

Statement II: On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O , it has higher boiling point.

In the light of the above statements, choose the *correct* answer from the options given below:

- Statement I is true but Statement II is false.
 - Statement I is false but Statement II is true.
 - Both Statement I and Statement II are true.
 - Both Statement I and Statement II are false.
13. Among Group 16 elements, which one does **NOT** show -2 oxidation state? (2024)
- Te
 - Po
 - O
 - Se
14. Amongst the given options, which of the following molecules/ion acts as a Lewis acid? (2023)
- OH^-
 - NH_3
 - H_2O
 - BF_3
15. Match List-I with List-II: (2023)

	List-I (Oxoacids of Sulphur)		List-II (Bonds of Sulphur)
A.	Peroxodisulphuric acid	I.	Two S-OH, Four S=O, One S-O-S
B.	Sulphuric acid	II.	Two S-OH, One S=O
C.	Pyrosulphuric acid	III.	Two S-OH, Four S=O, One S-O-O-S
D.	Sulphurous acid	IV.	Two S-OH, Two S=O

Choose the correct answer from the options given below :

- A-III, B-IV, C-II, D-I
 - A-I, B-III, C-II, D-IV
 - A-III, B-IV, C-I, D-II
 - A-I, B-III, C-IV, D-II
16. Given below are two statements: (2022)
- Statement-I:** The boiling points of the following hydrides of group 16 elements increases in the order-
 $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$.
- Statement-II:** The boiling points of these hydrides increase with increase in molar mass.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- Statement-I** is incorrect but **Statement-II** is correct.
 - Both **Statement-I** and **Statement-II** are correct
 - Both **Statement-I** and **Statement-II** are incorrect
 - Statement-I** is correct but **Statement-II** is incorrect
17. In which one of the following arrangements the given sequence is **not** strictly according to the properties indicated against it? (2021)
- $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing pK_a values
 - $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing acidic character
 - $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: Increasing oxidizing power
 - $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength

18. Which of the following oxoacid of sulphur has $-\text{O}-\text{O}-$ linkage? (2020)
- H_2SO_4 , sulphuric acid
 - $\text{H}_2\text{S}_2\text{O}_8$, peroxodisulphuric acid
 - $\text{H}_2\text{S}_2\text{O}_7$, pyrosulphuric acid
 - H_2SO_3 , sulphurous acid

19. The reaction of concentrated sulphuric acid with carbohydrates ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is an example of (2020-Covid)
- Oxidation
 - Reduction
 - Sulphonation
 - Dehydration
20. Which is the correct thermal stability order for H_2E (E = O, S, Se, Te and Po)? (2019)
- $\text{H}_2\text{S} < \text{H}_2\text{O} < \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{Po}$
 - $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{Po}$
 - $\text{H}_2\text{Po} < \text{H}_2\text{Te} < \text{H}_2\text{Se} < \text{H}_2\text{S} < \text{H}_2\text{O}$
 - $\text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{Po} < \text{H}_2\text{O} < \text{H}_2\text{S}$
21. In which pair of ions both the species contain S – S bond? (2017-Delhi)
- $\text{S}_4\text{O}_6^{2-}, \text{S}_2\text{O}_7^{2-}$
 - $\text{S}_2\text{O}_7^{2-}, \text{S}_2\text{O}_3^{2-}$
 - $\text{S}_4\text{O}_6^{2-}, \text{S}_2\text{O}_3^{2-}$
 - $\text{S}_2\text{O}_7^{2-}, \text{S}_2\text{O}_8^{2-}$
22. Which of the statements given below is incorrect? (2015 Re)
- O_3 molecule is bent
 - ONF is isoelectronic with O_2N^-
 - OF_2 is an oxide of fluorine
 - Cl_2O_7 is an anhydride of perchloric acid
23. Acidity of diprotic acids in aqueous solutions increases in the order: (2014)
- $\text{H}_2\text{Se} < \text{H}_2\text{S} < \text{H}_2\text{Te}$
 - $\text{H}_2\text{Te} < \text{H}_2\text{S} < \text{H}_2\text{Se}$
 - $\text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{S}$
 - $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$

Group 17 Elements (Halogens)

24. Identify the incorrect statement from the following: (2024 Re)
- The acidic strength of HX (X = F, Cl, Br and I) follows the order: $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$.
 - Fluorine exhibits -1 oxidation state whereas other halogens exhibit $+1, +3, +5$ and $+7$ oxidation states also.
 - The enthalpy of dissociation of F_2 is smaller than that of Cl_2 .
 - Fluorine is stronger oxidising agent than chlorine.
25. Fluorine is a stronger oxidising agent than chlorine because:
- F-F bond has a low enthalpy of dissociation.
 - Fluoride ion (F^-) has high hydration enthalpy.
 - Electron gain enthalpy of fluorine is less negative than chlorine.
 - Fluorine has a very small size.
- Choose the **most appropriate** answer from the options given: (2022 Re)
- (b) and (c) only
 - (a) and (b) only
 - (a) and (c) only
 - (a) and (d) only
26. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** (2022)
- Assertion (A):** ICl is **more** reactive than I_2 .
- Reason (R):** I-Cl bond is **weaker** than I-I bond
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (A) is not correct but (R) is correct.
 - Both (A) and (R) are correct and (R) is the correct explanation of (A).
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 - (A) is correct but (R) is not correct.

27. **Statement-I:** Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$. (2021)

Statement-II: As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

In the light of the above statements, choose the correct answer from the options given below.

- a. Both Statement-I and Statement-II are false
b. Statement-I is correct but Statement-II is false
c. Statement-I is incorrect but Statement-II is true
d. Both Statement-I and Statement-II are true
28. Which one of the following reactions does not come under hydrolysis type reaction? (2020-Covid)
- a. $\text{Li}_3\text{N(s)} + 3\text{H}_2\text{O(l)} \rightarrow \text{NH}_3\text{(s)} + 3\text{LiOH(aq)}$
b. $2\text{F}_2\text{(g)} + 2\text{H}_2\text{O(l)} \rightarrow 4\text{HF(aq)} + \text{O}_2\text{(g)}$
c. $\text{P}_4\text{O}_{10}\text{(s)} + 6\text{H}_2\text{O(l)} \rightarrow 4\text{H}_3\text{PO}_4\text{(aq)}$
d. $\text{SiCl}_4\text{(l)} + 2\text{H}_2\text{O(l)} \rightarrow \text{SiO}_2\text{(s)} + 4\text{HCl(aq)}$
29. Which of the following statements is not true for halogens? (2018)
- a. All form monobasic oxyacids
b. All are oxidizing agents
c. Chlorine has the highest electron-gain enthalpy
d. All but fluorine show positive oxidation states
30. In the structure of ClF_3 , the number of lone pair of electrons on central atom 'Cl' is (2018)
- a. One b. Two c. Three d. Four
31. Match the interhalogen compounds of Column-I with the geometry in Column-II and assign the correct code: (2017-Delhi)

	Column-I		Column-II
(A)	XX'	(i)	T-shape
(B)	XX'_3	(ii)	Pentagonal bipyramidal
(C)	XX'_5	(iii)	Linear
(D)	XX'_7	(iv)	Square-pyramidal
		(v)	Tetrahedral

Code:

- (A) (B) (C) (D) (A) (B) (C) (D)
a. (iv) (iii) (ii) (i) b. (iii) (iv) (i) (ii)
c. (iii) (i) (iv) (ii) d. (v) (iv) (iii) (ii)
32. Which one of the following orders is correct for the bond dissociation enthalpy of halogen molecules? (2016 - I)
- a. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ b. $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
c. $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$ d. $\text{Br}_2 > \text{I}_2 > \text{F}_2 > \text{Cl}_2$
33. Among the following, the correct order of acidity is: (2016 - I)
- a. $\text{HClO}_4 < \text{HClO}_2 < \text{HClO} < \text{HClO}_3$
b. $\text{HClO}_3 < \text{HClO}_4 < \text{HClO}_2 < \text{HClO}$
c. $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$
d. $\text{HClO}_2 < \text{HClO} < \text{HClO}_3 < \text{HClO}_4$

34. The variation of the boiling points of the hydrogen halides is in the order $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$. What explains the higher boiling point of hydrogen fluoride? (2015 Re)

- a. The effect of nuclear shielding is much reduced in fluorine which polarizes the HF molecule
b. The electronegativity of fluorine is much higher than for other elements in the group
c. There is strong hydrogen bonding between HF molecules
d. The bond energy of HF molecules is greater than in other hydrogen halides

35. Which is the strongest acid in the following? (2013)

- a. H_2SO_4 b. HClO_3 c. HClO_4 d. H_2SO_3

Group 18 Elements (Noble Gases)

36. Match List-I with List-II : (2024 Re)

	List-I (Atom/Molecule)		List-II (Property)
A.	Nitrogen atom	I.	Paramagnetic
B.	Fluorine molecule	II.	Most reactive element in group 18
C.	Oxygen molecule	III.	Element with highest ionisation enthalpy in group 15
D.	Xenon atom	IV.	Strongest oxidising agent

Identify the correct answer from the options given below:

- a. A-III, B-I, C-IV, D-II b. A-I, B-IV, C-III, D-II
c. A-II, B-IV, C-I, D-III d. A-III, B-IV, C-I, D-II

37. Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them. (2021)

- a. Noble gases have very high melting and boiling points
b. Noble gases have weak dispersion forces
c. Noble gases have large positive values of electron gain enthalpy
d. Noble gases are sparingly soluble in water

38. Match the compounds of Xe in Column-I with the molecular structure in Column-II. (2020-Covid)

	Column-I		Column-II
(A)	XeF_2	(i)	Square planar
(B)	XeF_4	(ii)	Linear
(C)	XeO_3	(iii)	Square pyramidal
(D)	XeOF_4	(iv)	Pyramidal

- a. A-ii B-i C-iv D-iii
b. A-ii B-i C-iii D-iv
c. A-ii B-iv C-iii D-i
d. A-ii B-iii C-i D-iv

39. Match the Xenon compounds in Column-I with its structure in Column-II and assign the correct code: (2019)

	Column-I		Column-II
(A)	XeF_4	(i)	Pyramidal
(B)	XeF_6	(ii)	Square planar
(C)	XeOF_4	(iii)	Distorted octahedral
(D)	XeO_3	(iv)	Square pyramidal

Code:

- | | | | | | | | |
|-------|-----|-----|----|--------|-----|----|----|
| A | B | C | D | A | B | C | D |
| a. i | ii | iii | iv | b. ii | iii | iv | i |
| c. ii | iii | i | iv | d. iii | iv | i | ii |

40. Which of the following pairs of compounds is isoelectronic and isostructural? (2017-Delhi)

- | | |
|------------------------------------|--------------------------------------|
| a. IF_3 , XeF_2 | b. BeCl_2 , XeF_2 |
| c. TeI_2 , XeF_2 | d. IBr_2^- , XeF_2 |

41. The correct geometry and hybridization for XeF_4 are: (2016 - II)

- a. Planar triangle, sp^3d^3
 b. Square planar, sp^3d^2
 c. Octahedral, sp^3d^2
 d. Trigonal bipyramidal, sp^3d

42. Match the compound given in Column-I with the hybridization and shape given in Column-II and mark the correct option. (2016 - I)

	Column-I		Column-II
(A)	XeF_6	(i)	Distorted octahedral
(B)	XeO_3	(ii)	Square planar
(C)	XeOF_4	(iii)	Pyramidal
(D)	XeF_4	(iv)	Square pyramidal

Code:

- | | | | | | | | |
|---------|------|------|-------|---------|-------|------|------|
| (A) | (B) | (C) | (D) | (A) | (B) | (C) | (D) |
| a. (iv) | (i) | (ii) | (iii) | b. (i) | (iii) | (iv) | (ii) |
| c. (i) | (ii) | (iv) | (iii) | d. (iv) | (iii) | (i) | (ii) |

43. In which of the following pairs, both the species are not isostructural? (2015 Re)

- a. XeF_4 , XeO_4
 b. SiCl_4 , PCl_4^+
 c. Diamond, silicon carbide
 d. NH_3 , PH_3

44. XeF_2 is isostructural with: (2013)

- a. TeF_2 b. ICl_2^- c. SbCl_3 d. BaCl_2

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (c) | 3. (a) | 4. (c) | 5. (d) | 6. (a) | 7. (b) | 8. (b) | 9. (b) | 10. (b) |
| 11. (d) | 12. (c) | 13. (b) | 14. (d) | 15. (c) | 16. (c) | 17. (a) | 18. (b) | 19. (d) | 20. (c) |
| 21. (c) | 22. (c) | 23. (d) | 24. (a) | 25. (b) | 26. (b) | 27. (d) | 28. (b) | 29. (d) | 30. (b) |
| 31. (c) | 32. (c) | 33. (c) | 34. (c) | 35. (c) | 36. (d) | 37. (a) | 38. (a) | 39. (b) | 40. (d) |
| 41. (b) | 42. (b) | 43. (a) | 44. (b) | | | | | | |